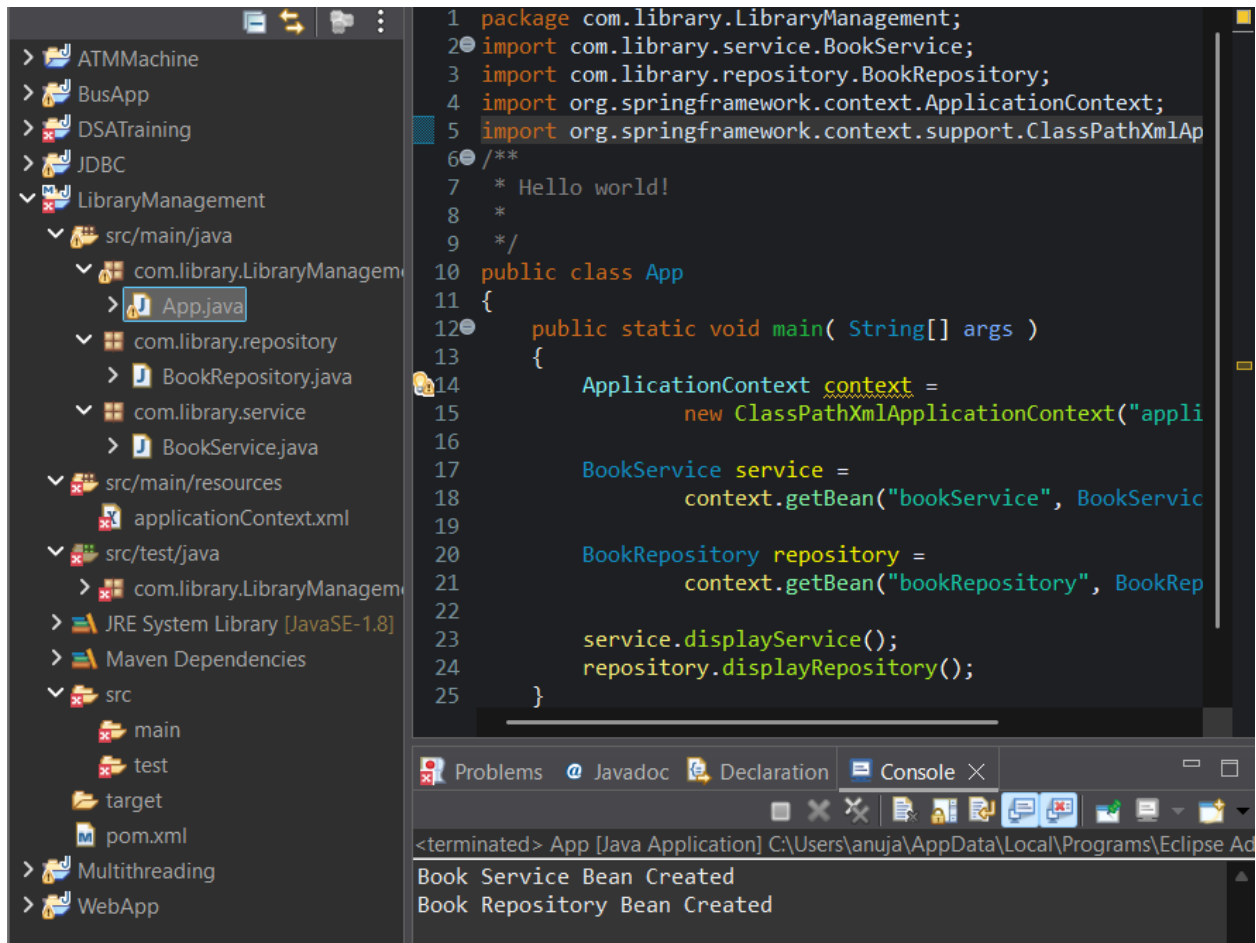


Exercise 1: Configuring a Basic Spring Application

Scenario:

Your company is developing a web application for managing a library. You need to use the Spring Framework to handle the backend operations.



Exercise 2: Implementing Dependency Injection

Scenario:

In the library management application, you need to manage the dependencies between the BookService and BookRepository classes using Spring's IoC and DI.

Code:

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {
```

```

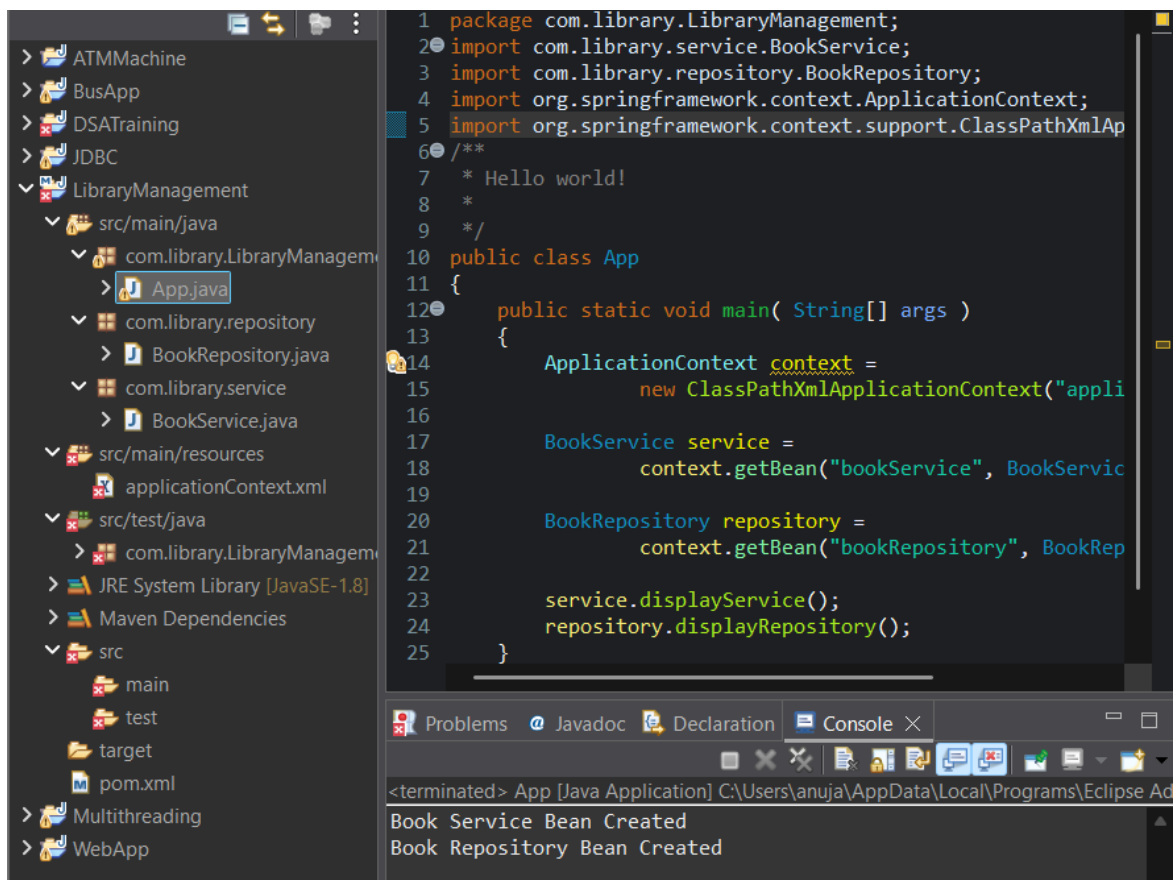
private BookRepository bookRepository;

// Setter for Dependency Injection
public void setBookRepository(BookRepository bookRepository) {
    this.bookRepository = bookRepository;
}

public void displayService() {
    System.out.println("Book Service Bean Created");

    // Call repository method
    bookRepository.displayRepository();
}
}

```

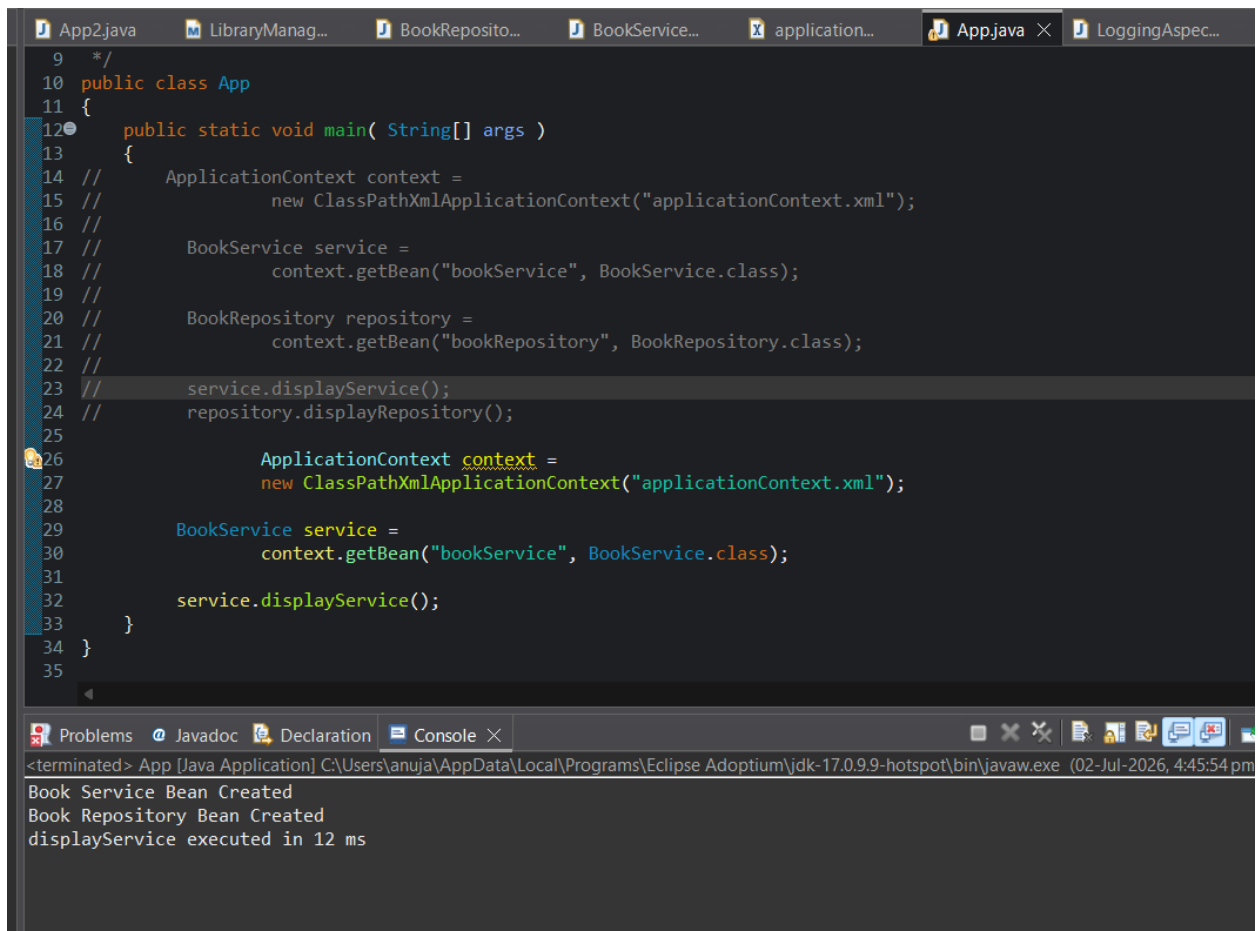


```
<terminated> App [Java Application] C:\Users\anuja\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.9-hotspot\bin\javaw.exe
Book Service Bean Created
Book Repository Bean Created
```

Exercise 3: Implementing Logging with Spring AOP

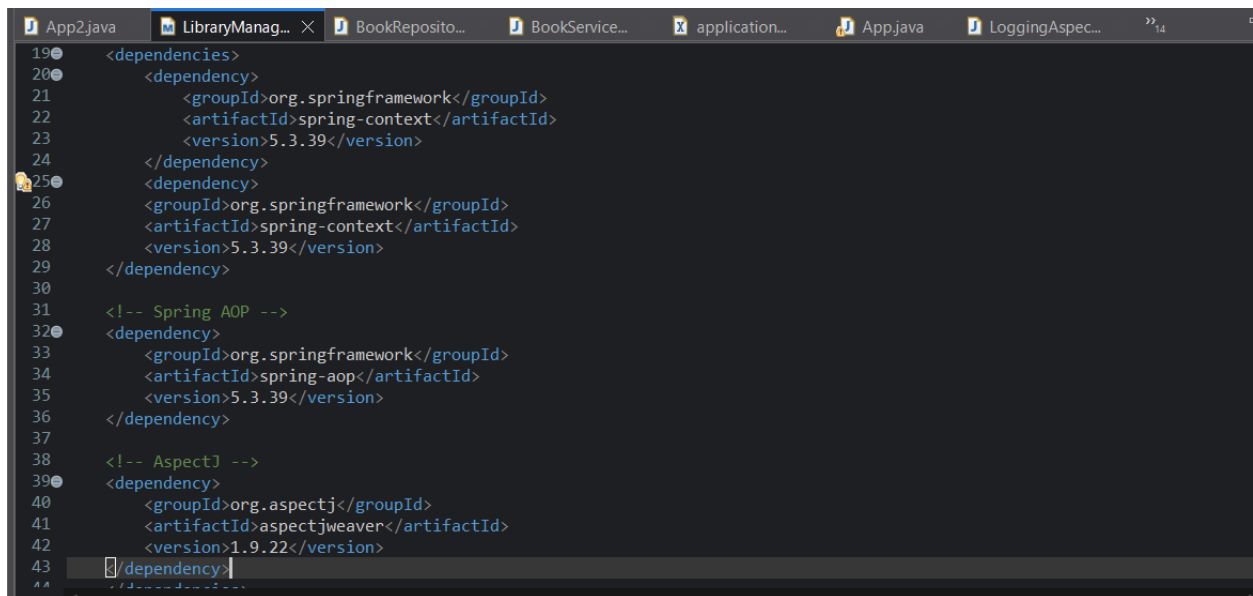
Scenario:

The library management application requires logging capabilities to track method execution times.



```
App2.java LibraryManag... BookReposito... BookService... application... App.java X LoggingAspec...
9  */
10 public class App
11 {
12     public static void main( String[] args )
13     {
14         // ApplicationContext context =
15         //     new ClassPathXmlApplicationContext("applicationContext.xml");
16         //
17         // BookService service =
18         //     context.getBean("bookService", BookService.class);
19         //
20         // BookRepository repository =
21         //     context.getBean("bookRepository", BookRepository.class);
22         //
23         // service.displayService();
24         // repository.displayRepository();
25
26         ApplicationContext context =
27             new ClassPathXmlApplicationContext("applicationContext.xml");
28
29         BookService service =
30             context.getBean("bookService", BookService.class);
31
32         service.displayService();
33     }
34 }
35
```

```
Problems Javadoc Declaration Console X
<terminated> App [Java Application] C:\Users\anuja\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.9-hotspot\bin\javaw.exe (02-Jul-2026, 4:45:54 pm)
Book Service Bean Created
Book Repository Bean Created
displayService executed in 12 ms
```

A screenshot of an IDE window showing a Maven pom.xml file. The file is named 'LibraryManag...' and is part of a project with several other files open in the background. The code defines a set of dependencies for a Spring-based application. It includes Spring Framework 5.3.39, Spring AOP 5.3.39, and AspectJ 1.9.22. The code is as follows:

```
19 <dependencies>
20 <dependency>
21   <groupId>org.springframework</groupId>
22   <artifactId>spring-context</artifactId>
23   <version>5.3.39</version>
24 </dependency>
25 <dependency>
26   <groupId>org.springframework</groupId>
27   <artifactId>spring-context</artifactId>
28   <version>5.3.39</version>
29 </dependency>
30
31 <!-- Spring AOP -->
32 <dependency>
33   <groupId>org.springframework</groupId>
34   <artifactId>spring-aop</artifactId>
35   <version>5.3.39</version>
36 </dependency>
37
38 <!-- AspectJ -->
39 <dependency>
40   <groupId>org.aspectj</groupId>
41   <artifactId>aspectjweaver</artifactId>
42   <version>1.9.22</version>
43 </dependency>
```

Exercise 4: Creating and Configuring a Maven Project

Scenario:

You need to set up a new Maven project for the library management application and add Spring dependencies.

The screenshot displays the Eclipse IDE interface. The top editor shows the `pom.xml` file with the following XML content:

```

19<dependencies>
20<dependency>
21<groupId>org.springframework</groupId>
22<artifactId>spring-context</artifactId>
23<version>5.3.39</version>
24</dependency>
25<dependency>
26<groupId>org.springframework</groupId>
27<artifactId>spring-context</artifactId>
28<version>5.3.39</version>
29</dependency>
30
31<!-- Spring AOP -->
32<dependency>
33<groupId>org.springframework</groupId>
34<artifactId>spring-aop</artifactId>
35<version>5.3.39</version>
36</dependency>
37
38<!-- AspectJ -->
39<dependency>
40<groupId>org.aspectj</groupId>
41<artifactId>aspectjweaver</artifactId>
42<version>1.9.22</version>
43</dependency>
44</dependencies>

```

The left sidebar shows the project structure:

- JDBC
 - LibraryManagement
 - src/main/java
 - com.library.aspect
 - LoggingAspect.java
 - com.library.LibraryManagement
 - App.java
 - App
 - main(String[]) : void
 - com.library.repository
 - BookRepository.java
 - com.library.service
 - BookService.java
 - src/main/resources
 - applicationContext.xml
 - src/test/java
 - com.library.LibraryManagement
 - JRE System Library [JavaSE-1.8]
 - Maven Dependencies
 - aspectjweaver-1.9.22.jar - C:\Users\anuja\.m2\repository\org\aspectj
 - spring-aop-5.3.39.jar - C:\Users\anuja\.m2\repository\org\springfram
 - spring-beans-5.3.39.jar - C:\Users\anuja\.m2\repository\org\springfra
 - spring-context-5.3.39.jar - C:\Users\anuja\.m2\repository\org\springfra
 - spring-core-5.3.39.jar - C:\Users\anuja\.m2\repository\org\springfran
 - spring-expression-5.3.39.jar - C:\Users\anuja\.m2\repository\org\sprii
 - spring-jcl-5.3.39.jar - C:\Users\anuja\.m2\repository\org\springframe

The bottom editor shows the `Console` output:

```

<terminated> App [Java Application] C:\Users\anuja\AppData\Local\Programs\Eclipse Adoptium\jdk
Book Service Bean Created
Book Repository Bean Created

```

Exercise 5: Configuring the Spring IoC Container

Scenario:

The library management application requires a central configuration for beans and dependencies.

```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xmlns:aop="http://www.springframework.org/schema/aop"
6       xsi:schemaLocation="
7       http://www.springframework.org/schema/beans
8       https://www.springframework.org/schema/beans/spring-beans.xsd
9       http://www.springframework.org/schema/aop
10      https://www.springframework.org/schema/aop/spring-aop.xsd">
11
12   <bean id="bookRepository"
13         class="com.library.repository.BookRepository"/>
14
15   <bean id="bookService"
16         class="com.library.service.BookService">
17
18     <property name="bookRepository" ref="bookRepository"/>
19   </bean>
20
21   <bean id="loggingAspect"
22         class="com.library.aspect.LoggingAspect"/>
23
24   <aop:aspectj-autoproxy/>
25
26 </beans>
```

Problems Javadoc Declaration Console X

<terminated> App [Java Application] C:\Users\anuja\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.9-hotspot\bin\javaw.e

Book Service Bean Created
Book Repository Bean Created
displayService executed in 12 ms

Exercise 6: Configuring Beans with Annotations

Scenario:

You need to simplify the configuration of beans in the library management application using annotations.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xmlns:aop="http://www.springframework.org/schema/aop"
6       xsi:schemaLocation="
7         http://www.springframework.org/schema/beans
8         https://www.springframework.org/schema/beans/spring-beans.xsd
9         http://www.springframework.org/schema/aop
10        https://www.springframework.org/schema/aop/spring-aop.xsd">
11
12   <context:component-scan base-package="com.library.LibraryManagement"/>
13
14   </beans>

```

App2.java LibraryManag... BookRepository... BookService... applica...

Maximize

```

1 package com.library.service;
2
3 import org.springframework.beans.factory.annotation.Autowired;
4 import org.springframework.stereotype.Service;
5
6 import com.library.repository.BookRepository;
7
8 @Service
9 public class BookService {
10
11     @Autowired
12     private BookRepository bookRepository;
13
14     public void displayService() {
15         System.out.println("Book Service Bean Created");
16         bookRepository.displayRepository();
17     }
18 }

```

```
1 package com.library.repository;
2
3 import org.springframework.stereotype.Repository;
4
5 @Repository
6 public class BookRepository {
7
8     public void displayRepository() {
9         System.out.println("Book Repository Bean Created");
10    }
11 }
```

Exercise 7: Implementing Constructor and Setter Injection

Scenario:

The library management application requires both constructor and setter injection for better control over bean initialization.


```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6       http://www.springframework.org/schema/beans
7       https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9   <bean id="library"
10       class="com.library.repository.Library"/>
11
12   <bean id="bookRepository"
13       class="com.library.repository.BookRepository"/>
14
15   <bean id="bookService"
16       class="com.library.service.BookService">
17
18       <!-- Constructor Injection -->
19       <constructor-arg ref="library"/>
20
21       <!-- Setter Injection -->
22       <property name="bookRepository"
23               ref="bookRepository"/>
24   </bean>
25 </beans>
```

Problems Javadoc Declaration Console X

<terminated> App [Java Application] C:\Users\anuja\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.9-hotspot\bin\javaw.exe (02-Jul-2023 10:00:00 AM)

Book Service Bean Created
Library Bean Created
Book Repository Bean Created

Exercise 8: Implementing Basic AOP with Spring

Scenario:

The library management application requires basic AOP functionality to separate cross-cutting concerns like logging and transaction management.

```

1 package com.library.LibraryManagement;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 import com.library.service.BookService;
7
8 public class App {
9
10     public static void main(String[] args) {
11
12         ApplicationContext context =
13             new ClassPathXmlApplicationContext("applicationContext.xml");
14
15         BookService service =
16             context.getBean(BookService.class);
17
18         service.displayService();
19     }
20 }

```

Problems Javadoc Declaration Console ×

<terminated> App [Java Application] C:\Users\anuja\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.9-hotspot\bin\javaw.exe (02-Jul-2024 10:00:00 AM)

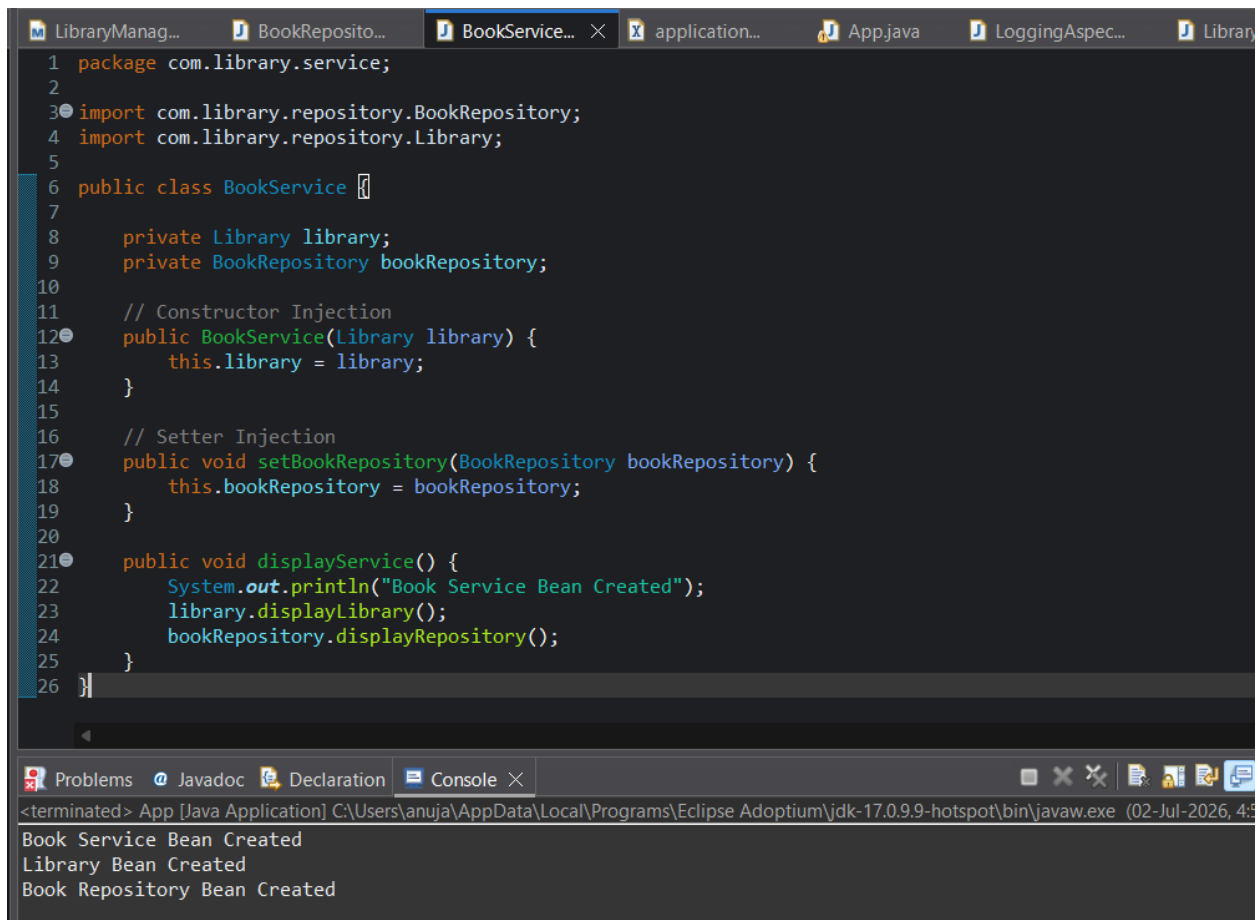
Before method execution
 Book Service Bean Created
 Book Repository Bean Created
 After method execution

```
1 package com.library.aspect;
2
3 import org.aspectj.lang.annotation.After;
4 import org.aspectj.lang.annotation.Aspect;
5 import org.aspectj.lang.annotation.Before;
6
7 @Aspect
8 public class LoggingAspect {
9
10     @Before("execution(* com.library.service.*(..))")
11     public void beforeMethod() {
12         System.out.println("Before method execution");
13     }
14
15     @After("execution(* com.library.service.*(..))")
16     public void afterMethod() {
17         System.out.println("After method execution");
18     }
19 }
```

Problems Javadoc Declaration Console X

<terminated> App [Java Application] C:\Users\anuja\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.9-hotspot\bin\javaw

Before method execution
Book Service Bean Created
Book Repository Bean Created
After method execution



The screenshot shows the Eclipse IDE with several tabs open: LibraryManag..., BookRepository..., BookService..., application..., App.java, LoggingAspec..., and Library. The 'BookService...' tab is active, displaying the following Java code:

```
1 package com.library.service;
2
3 import com.library.repository.BookRepository;
4 import com.library.repository.Library;
5
6 public class BookService {
7
8     private Library library;
9     private BookRepository bookRepository;
10
11     // Constructor Injection
12     public BookService(Library library) {
13         this.library = library;
14     }
15
16     // Setter Injection
17     public void setBookRepository(BookRepository bookRepository) {
18         this.bookRepository = bookRepository;
19     }
20
21     public void displayService() {
22         System.out.println("Book Service Bean Created");
23         library.displayLibrary();
24         bookRepository.displayRepository();
25     }
26 }
```

Below the code editor, the 'Console' tab is active, showing the output of the application:

```
<terminated> App [Java Application] C:\Users\anuja\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.9-hotspot\bin\javaw.exe (02-Jul-2026, 4:5
Book Service Bean Created
Library Bean Created
Book Repository Bean Created
```

Exercise 9: Creating a Spring Boot Application

Scenario:

You need to create a Spring Boot application for the library management system to simplify configuration and deployment.

BookService... application... App.java application... BookControll... Bookjava BookRepo.java LibraryManag... X 19

```
1 package com.library.LibraryManagement;
2
3 import org.springframework.boot.SpringApplication;
4 @SpringBootApplication
5 public class LibraryManagementApplication {
6
7     public static void main(String[] args) {
8         SpringApplication.run(LibraryManagementApplication.class, args);
9     }
10 }
11
12
13 }
```

Problems Javadoc Declaration Console X

LibraryManagementApplication [Java Application] C:\Users\anjuja\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.9-hotspot\bin\javaw.exe (02-Jul-2026, 5:21:31 pm elapsed 0:00:35)

Spring Boot (v3.5.16)

```
2026-07-02T17:21:33.281+05:30 INFO 20656 --- [main] c.l.l.LibraryManagementApplication : Starting LibraryManagementAppl
2026-07-02T17:21:33.285+05:30 INFO 20656 --- [main] c.l.l.LibraryManagementApplication : No active profile set, falling
2026-07-02T17:21:34.015+05:30 INFO 20656 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA
2026-07-02T17:21:34.089+05:30 INFO 20656 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository
2026-07-02T17:21:34.745+05:30 INFO 20656 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 80
2026-07-02T17:21:34.762+05:30 INFO 20656 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-02T17:21:34.762+05:30 INFO 20656 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apac
2026-07-02T17:21:34.828+05:30 INFO 20656 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded W
2026-07-02T17:21:34.830+05:30 INFO 20656 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: in
2026-07-02T17:21:35.037+05:30 INFO 20656 --- [main] o.hibernate.jpa.internal.util.LogHelper : HHH000204: Processing Persiste
2026-07-02T17:21:35.105+05:30 INFO 20656 --- [main] org.hibernate.Version : HHH0000412: Hibernate ORM core vers
2026-07-02T17:21:35.153+05:30 INFO 20656 --- [main] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache
2026-07-02T17:21:35.473+05:30 INFO 20656 --- [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignor
2026-07-02T17:21:35.517+05:30 INFO 20656 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-07-02T17:21:35.827+05:30 INFO 20656 --- [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection
2026-07-02T17:21:35.830+05:30 INFO 20656 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed
2026-07-02T17:21:35.905+05:30 INFO 20656 --- [main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
```

ATMMachine
BusApp
DSATraining
Ex9
src/main/java
com.library.LibraryManagement
LibraryManagementApplication.java
com.library.LibraryManagement.controller
BookController.java
com.library.LibraryManagement.entity
Bookjava
com.library.LibraryManagement.repository
BookRepo.java
src/main/resources
static
templates
application.properties
src/test/java
JRE System Library [Javase-17]
Maven Dependencies
src
target
HELP.md
mvnw
mvnw.cmd
pom.xml
JDBC
LibraryManagement
Multithreading
WebApp

```
1 package com.library.LibraryManagement;
2
3 import org.springframework.boot.SpringApplication;
4 @SpringBootApplication
5 public class LibraryManagementApplication {
6
7     public static void main(String[] args) {
8         SpringApplication.run(LibraryManagementApplication.class, args);
9     }
10 }
11
12
13 }
```

Problems Javadoc Declaration Console X

LibraryManagementApplication [Java Application] C:\Users\anjuja\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.9-hotspot\bin\javaw.exe (02-Jul-2026, 5:21:31 pm elapsed 0:00:42) [pid: 2

```
2026-07-02T17:21:34.762+05:30 INFO 20656 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-02T17:21:34.762+05:30 INFO 20656 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache
2026-07-02T17:21:34.828+05:30 INFO 20656 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebAg
2026-07-02T17:21:34.830+05:30 INFO 20656 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initia
2026-07-02T17:21:35.037+05:30 INFO 20656 --- [main] o.hibernate.jpa.internal.util.LogHelper : HHH000204: Processing Persistencel
2026-07-02T17:21:35.105+05:30 INFO 20656 --- [main] org.hibernate.Version : HHH0000412: Hibernate ORM core vers
2026-07-02T17:21:35.153+05:30 INFO 20656 --- [main] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache diss
2026-07-02T17:21:35.473+05:30 INFO 20656 --- [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring
2026-07-02T17:21:35.517+05:30 INFO 20656 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-07-02T17:21:35.827+05:30 INFO 20656 --- [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection co
2026-07-02T17:21:35.830+05:30 INFO 20656 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-07-02T17:21:35.905+05:30 INFO 20656 --- [main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [Connecting through datasource 'HikariDataSource (HikariPool-1)']
Database driver: undefined/unknown
Database version: 2.3.232
Autocommit mode: undefined/unknown
Isolation level: undefined/unknown
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-07-02T17:21:36.673+05:30 INFO 20656 --- [main] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH000489: No JTA platform availa
Hibernate: create table book (id bigint generated by default as identity, author varchar(255), title varchar(255), primary key (id))
2026-07-02T17:21:36.723+05:30 INFO 20656 --- [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFacto
2026-07-02T17:21:36.940+05:30 WARN 20656 --- [main] jpabase.Configuration$JpaWebConfiguration : spring.jpa.open-in-view is enable
2026-07-02T17:21:37.291+05:30 INFO 20656 --- [main] o.s.b.a.h2.H2ConsoleAutoConfiguration : H2 console available at '/h2-conse
2026-07-02T17:21:37.422+05:30 INFO 20656 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http)
2026-07-02T17:21:37.438+05:30 INFO 20656 --- [main] c.l.l.LibraryManagementApplication : Started LibraryManagementApplicati
```